

Compositional Operator Networks: 15-Week Project Plan

A semester of structured loss-ablation studies on FlowBench and a 2D Boussinesq warm-up

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Postdoc + 2 undergraduate research assistants

Light involvement: A. Sharma group (Iowa State Aerospace)

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Summary

This document scopes a 15-week empirical programme that (i) brings the compositional operator-network framework onto FlowBench data and a 2D Rayleigh–Bénard warm-up, (ii) characterises the contribution of each loss family by controlled ablation, (iii) maps Pareto trade-offs across loss configurations, and (iv) produces the empirical content for the anchor paper. The plan is structured around four phases, weekly milestones, and clear role separation between the postdoc (framework owner), two undergraduate research assistants (one ML-leaning, one physics-leaning), and the PI. Sharma’s involvement is light during this semester and ramps toward Paper 2 in the following one.

Companion documents. (a) *Compositional Operator Networks for Parametric, Geometry-Dependent Flow Fields* — the anchor paper draft, the scientific scope of the 15 weeks. (b) *Compositional Operator Framework* — the long-form technical document with the full loss library and architecture survey.

1 Goals and success criteria

The 15 weeks have one primary scientific goal and three subsidiary goals.

Primary goal. Quantify, by controlled ablation, how each loss family in the framework contributes to (a) field reconstruction quality, (b) latent-space identifiability as measured by linear probing and concept-vector arithmetic, and (c) downstream physical metrics (Strouhal and Nusselt for Case 1, amplitude-equation recovery for the warm-up).

Subsidiary goals.

1. Establish a working pipeline and codebase with reproducible training and evaluation on at least one standard FlowBench subset and the 2D Boussinesq warm-up.
2. Produce the diagnostic suite (linear probes, concept-vector arithmetic matrices, swap-consistency tests, spectral comparisons) as a standalone evaluation tool that can be re-used on Cases 2 and 3 next semester.
3. Draft the empirical sections of the anchor paper, with all figures, tables, and ablation results in publication-ready form.

Success criteria (quantitative, by Week 15).

1. Best-configuration relative L^2 reconstruction error on FlowBench-derived test set comparable to or better than the FlowBench baselines reported in [Tali et al. \(2025\)](#).
2. Linear-probe $R^2 \geq 0.85$ on at least three operating/regime parameters from z_μ on the warm-up dataset (a sanity check on basic identifiability).
3. A concept-vector disentanglement matrix on Case 1 with diagonal entries $\geq 0.9\sigma/\gamma$ and off-diagonal entries $\leq 0.3\sigma/\gamma$ on the principal axes (matching the AeroJEPA ([AeroJEPA, 2026](#)) pattern).
4. Pareto front mapped across at least three loss configurations with weight sweeps documented.
5. Anchor paper draft complete with empirical Sections 5 and 7 fully populated.

2 Team and roles

Role	Time	Responsibilities
PI (Ganapathysubramanian)	~2–3 hr/wk	Scientific direction, weekly review, paper writing in Phase 4, coordination with Sharma group, hand-off planning.
Postdoc	~30–35 hr/wk	Framework codebase ownership; training-loop implementation; compositional losses (L 10–12) which require group-batched data infrastructure; adaptive weighting (Grad-Norm, MGDA); FlowBench-pipeline integration; coordination of weekly milestones with both undergrads.
Undergrad A (ML-leaning)	~15–20 hr/wk	Diagnostic suite implementation: linear probes, concept-vector arithmetic, ridge probes for design-parameter recovery. Disentanglement losses (L 6–8). Pareto-front visualisation in Phase 3. Anchor-paper figure generation in Phase 4.
Undergrad B (physics-leaning)	~15–20 hr/wk	2D Boussinesq solver and warm-up dataset generation. Physics-informed losses (L 18–19) and spectral losses (L 22–23). SINDy-on-latents experiment for amplitude-equation recovery. Strouhal/Nusselt calculation pipeline from decoded fields.
A. Sharma (collaborator, ISU Aero)	~1 hr/wk (light)	Advisory on dynamic-stall framing in advance of Paper 2; review of Case 3 plan in Week 12–13; provision of pitching-airfoil dataset access for next-semester ramp-up.

The two undergraduates work in parallel on related but non-blocking tracks. Each track produces a self-contained deliverable that the postdoc integrates into the master codebase.

3 Phase plan

The 15 weeks are divided into four phases. Each phase has clear entry and exit conditions, with a one-week buffer baked into Phase 3 for catch-up and unanticipated issues.

Phase	Weeks	Focus	Exit condition
Phase 1	1–3	Foundation and baseline	Working training loop; baseline reconstruction; diagnostic pipeline tested.
Phase 2	4–9	Loss-family ablation	All within-family ablations complete; best-of-family configurations identified.
Phase 3	10–12	Pareto and adaptive weighting	Pareto front mapped; adaptive vs static comparison done.
Phase 4	13–15	Synthesis and writing	Anchor paper draft complete with all figures.

3.1 Phase 1 (Weeks 1–3): Foundation

The goal of Phase 1 is a working pipeline with a reconstruction-only baseline, on both the warm-up and a FlowBench subset, plus the diagnostic suite tested on the baseline output. Phase 1 gates Phase 2: no ablation studies until the baseline is reproducible.

Week 1 — Kickoff.

Team meeting to align on goals, codebase, and shared conventions. Postdoc clones the framework codebase (built on existing operator-network infrastructure from Rabeh et al., 2025b; Karki et al., 2025) and validates it on a small FlowBench subset. Undergrad A reads AeroJEPA (2026) for the diagnostic methodology and starts implementing linear probes against synthetic latent vectors. Undergrad B sets up an off-the-shelf pseudo-spectral 2D Boussinesq solver (e.g., Dedalus) and generates a small pilot dataset.

Week 2 — Pipeline assembly.

Postdoc has full training loop running with $\mathcal{L}_{\text{rec}}$ (L1) only, on FlowBench subset; logs reconstruction metrics. Undergrad A has linear-probe pipeline accepting trained latents and producing R^2 tables and concept-vector arithmetic matrices on synthetic data. Undergrad B has the warm-up dataset complete: a ~ 20 -point grid in $(\text{Ra}, \text{Pr}, \Gamma)$ with 5–10 realisations per grid point.

Week 3 — Baseline established.

Both datasets have reconstruction-only baselines. Diagnostic suite produces the first end-to-end report: reconstruction quality, z_μ probe R^2 on regime parameters, z_g probe R^2 on geometry parameters, baseline concept-vector arithmetic matrices. Team review meeting at end of Week 3: *exit gate to Phase 2*.

Phase 1 deliverables: (i) Reproducible training/evaluation codebase committed to a shared repo with branching strategy; (ii) baseline performance report; (iii) diagnostic-suite tooling.

3.2 Phase 2 (Weeks 4–9): Loss-family ablation

The core empirical phase. Six weeks divided into three sub-phases of two weeks each, one per loss family. Within each sub-phase, the team runs a small set of controlled experiments comparing successive additions of loss components to a fixed baseline. Each sub-phase ends with a within-family winner that is carried forward as the new baseline for the next sub-phase.

Sub-phase 2A (Weeks 4–5): Disentanglement family. Hypothesis: explicit decorrelation losses between latent blocks improve linear-probe R^2 on regime and geometry parameters at modest cost to reconstruction. Six configurations:

Configuration	Active losses	Owners
2A-baseline	L 1 only	Postdoc
2A-1	L 1 + L 6 (Pearson decorrelation)	Postdoc + Undergrad A
2A-2	L 1 + L 7 (HSIC)	Undergrad A
2A-3	L 1 + L 8 (block-Gram)	Postdoc
2A-4	L 1 + L 6 + L 8	Postdoc + Undergrad A
2A-5	L 1 + L 6 + L 7 + L 8	Postdoc + Undergrad A

Sub-phase 2B (Weeks 6–7): Compositional identifiability family. Hypothesis: group-structured supervision losses make the factorisation *functional*, not merely statistical, as measured by swap-consistency error. Requires implementing group-batched data loading. Five configurations:

Configuration	Active losses	Owners
2B-baseline	best from 2A	—
2B-1	2A-best + L 10 (same-factor invariance)	Postdoc
2B-2	2A-best + L 11 (contrastive, InfoNCE)	Postdoc + Undergrad A
2B-3	2A-best + L 12 (swap consistency)	Postdoc
2B-4	2A-best + L 10 + L 12	Postdoc

Sub-phase 2C (Weeks 8–9): Physics and spectral families. Hypothesis: physics-informed and spectral-statistical losses improve out-of-distribution generalisation and small-scale fidelity, particularly on regime extrapolation. Five configurations:

Configuration	Active losses	Owners
2C-baseline	best from 2B	—
2C-1	2B-best + L 18 (PDE residual)	Postdoc + Undergrad B
2C-2	2B-best + L 19 (conservation)	Undergrad B
2C-3	2B-best + L 22 (PSD matching, log-space)	Undergrad B
2C-4	2B-best + L 18 + L 22	Postdoc + Undergrad B

Across the six weeks of Phase 2, the team will run roughly 16 controlled training experiments. At an estimated 8–12 hours per training run on a single GPU, this requires ~ 160 GPU-hours and is fully accommodated within typical campus or lab compute allocations.

Phase 2 deliverables: (i) Ablation tables for all three loss families with reconstruction, probe, and swap-consistency metrics; (ii) within-family winners identified; (iii) preliminary versions of all empirical figures for the anchor paper; (iv) a brief technical report (10 pages, internal) summarising findings.

3.3 Phase 3 (Weeks 10–12): Pareto front and adaptive weighting

The goal of Phase 3 is to move beyond fixed loss-weight choices by (i) sweeping weights to map the Pareto front of (reconstruction, identifiability) trade-offs and (ii) testing whether adaptive weighting (GradNorm, uncertainty weighting, MGDA) recovers comparable performance without manual tuning.

Week 10 — Static weight sweep.

Take the top three configurations from Phase 2. For each, sweep one or two principal weight ratios on a small grid (e.g., $\lambda_{\text{disentangle}}/\lambda_{\text{rec}}$ across five values, $\lambda_{\text{physics}}/\lambda_{\text{rec}}$ across three). Plot the Pareto front of reconstruction L^2 vs probe R^2 . Owners: Postdoc + Undergrad A.

Week 11 — Adaptive weighting.

Implement and test GradNorm (Chen et al., 2018), uncertainty weighting (Kendall et al., 2018), and MGDA (Désidéri, 2012) as alternatives to static weights. Compare each adaptive scheme against the best static configuration on Case 1. Owners: Postdoc.

Week 12 — Catch-up and analysis buffer.

Resolve any outstanding issues from Phases 2 and 3. Generate publication-quality figures from cleaned-up data. Internal team review of all results; final decision on which configurations to highlight in the paper.

Phase 3 deliverables: (i) Pareto-front visualisation; (ii) adaptive-vs-static comparison table; (iii) all anchor-paper empirical figures finalised.

3.4 Phase 4 (Weeks 13–15): Synthesis and writing**Week 13 — Diagnostic deep-dive.**

Apply the full diagnostic suite to the chosen final configurations: linear-probe recovery tables, concept-vector arithmetic matrices, swap-consistency error reports, spectral comparisons. Undergrad B leads the SINDy-on-latents amplitude-equation recovery experiment for the warm-up; if successful, this is one of the strongest results in the paper.

Week 14 — Paper draft assembly.

PI and postdoc lead the writing. Empirical sections of the anchor paper (Sections 5 and 7 in current numbering) are populated with figures, tables, and quantitative results. Undergrads contribute methods text and figure captions for the components they owned.

Week 15 — Polish and hand-off.

Internal review of the draft. Postdoc finalises the empirical narrative. Team plans the next semester’s ramp-up: Case 3 (pitching airfoils with Sharma) becomes the primary focus, with Case 2 (electroconvection) as a parallel track if capacity allows. Final hand-off meeting with Sharma’s group to align on Paper 2 scope.

Phase 4 deliverables: (i) Anchor paper draft, complete in all empirical sections; (ii) plan and team allocation for next semester.

4 Loss-ablation methodology**4.1 What we measure**

For each training run, the evaluation pipeline produces six categories of metrics on a held-out test set. Items marked * are the primary metrics tracked across ablations.

1. **Reconstruction quality***: relative L^2 error per channel, RMSE/max-GT per channel, derivative-norm error.
2. **Latent-block identifiability***: 5-fold cross-validated ridge-probe R^2 for each known (μ, g) component on the corresponding latent block; cross-block R^2 (which should be low) as a sanity check.

3. **Concept-vector arithmetic***: for each known parameter, the $K \times K$ disentanglement-response matrix (one panel per pair of latent direction and parameter readout) following the construction of AeroJEPA (2026); reported as diagonal-mean and off-diagonal-mean sensitivities.
4. **Swap consistency**: $\mathcal{L}_{\text{swap}}$ value computed on held-out cross-combinations (μ_i, g_j) .
5. **Physics conformity**: PDE-residual norm and conservation-residual norm computed on decoded fields at unseen query points.
6. **Downstream physics***: for Case 1, Strouhal and Nusselt numbers from decoded fields compared to ground truth; for the warm-up, recovery fidelity of amplitude equations via SINDy on z_η .

The starred metrics form the headline ablation table; the others are supporting diagnostics.

4.2 Experimental controls

To isolate the contribution of each loss component, the ablation studies hold constant: the architecture (a fixed encoder–predictor–decoder triple, specifications below), the training schedule (cosine warmup, AdamW with fixed learning rate and weight decay), the data split (fixed seed for the train/val/test partition), and the random initialisation seed (3 seeds per configuration to estimate variance). Loss weights for the family-defining terms (e.g., L1 always at $\lambda_{\text{rec}} = 1$) are held at the default values across Phase 2, with weight sweeps deferred to Phase 3.

4.3 Architecture specification (working choice)

- **Geometry encoder**: coordinate-based SDF encoder in the Geom-DeepONet style (He et al., 2024; Rabeh et al., 2025a), producing z_g via tokens aggregated by farthest-point sampling and message passing. For the warm-up, z_g collapses to a small MLP on aspect ratio.
- **Regime encoder**: small MLP on the operating-condition vector, producing z_μ .
- **Dynamics propagator**: transformer or FNO block taking $(z_\mu, z_g, z_\eta(t))$ and producing $z_\eta(t + \Delta t)$. For Case 1 (steady), this collapses to identity and is omitted.
- **Decoder**: INR-style implicit decoder with Fourier-feature inputs (Tancik et al., 2020), conditioned on z and queried at arbitrary spatial points.

The exact dimensions of each block ($L_\mu, L_g, L_\eta, L_\xi$) are determined at training time by cross-validation, with an Occam prior favouring smaller blocks until reconstruction quality degrades. The same architecture is used across all ablations.

5 Risk register

Risk	Sev.	Mitigation
Compute time scarcity	Med	Book GPU allocation in Week 1; budget 200 GPU-hours conservatively; reserve some Phase 3 weight-sweep runs for parallelism on a cluster.
Codebase startup time	Med	Lean on existing operator-network infrastructure from Rabeh et al. (2025b) ; Karki et al. (2025) ; do not re-implement from scratch; postdoc owns this risk.
Group-batched data loading harder than expected	Med	Phase 2B is the gate; if implementation slips, postdoc carries the swap loss into Phase 3 as the primary configuration to compare adaptive weighting against.
Loss interactions don't show clean separations	Low	Run multiple seeds; report variance; if no signal, the negative result is itself valuable for the paper.
Undergraduate availability slips (exam periods, etc.)	Med	Plan around mid-term and finals weeks; lower-priority tasks pushed to buffer in Week 12.
RBC solver instability or unexpected behaviour	Low	Use established codes (Dedalus, Nek5000); cross-check against published RBC datasets; this is a textbook problem.
Sharma data-format integration delays	Low	Treated as a Phase 4 / next-semester item; not on the critical path for this semester.
Negative result on a key ablation (e.g., L12 doesn't help)	Low	A negative result is publishable and informative; report it cleanly with discussion.

6 Communication and meetings

Weekly cadence.

- Monday morning: 30-minute team standup (PI optional, postdoc + both undergrads required). Each person reports last week's progress, this week's plan, and any blockers.
- Friday afternoon: 1-hour PI-postdoc 1:1 covering scientific direction, paper outline, and next-week plan.
- Bi-weekly Wednesday: 1-hour technical deep-dive on the current sub-phase, with both undergrads presenting their experimental results.

End-of-phase reviews. Phase 1 (end of Week 3), Phase 2 (end of Week 9), Phase 3 (end of Week 12) all have one-hour all-hands review meetings. PI plus Sharma optional for Phase 2/3 reviews. The Phase 1 review is a hard gate: no Phase 2 work begins until Phase 1 deliverables are accepted.

Documentation. All experimental configurations, results, and figures live in the shared repo. A running ablation table is maintained as the canonical record of what has been run. Each undergraduate keeps a one-page weekly summary (text only) of progress and blockers; postdoc consolidates these into the Friday 1:1 with PI.

7 Compute and tooling requirements

- GPU: 1–2 NVIDIA A100 or H100 equivalents, ideally accessible to all team members; budget ~200 GPU-hours over the 15 weeks.
- Storage: ~1 TB for FlowBench subsets, warm-up dataset, model checkpoints, and ablation outputs.
- Codebase: PyTorch + standard scientific stack (NumPy, SciPy, scikit-learn for ridge probes); pseudo-spectral solver for the warm-up (Dedalus or equivalent); plotting via Matplotlib + Seaborn; version control on a shared platform with branch protection on `main`.
- Documentation: shared paper draft on Overleaf (PI + postdoc edit access; undergrads contribute via pull requests to a shared repo of figures and tables).

8 Hand-off plan to next semester

Outputs that carry forward.

- The framework codebase, with the diagnostic suite and the best-of-Phase-2 loss configuration as the default starting point for new cases.
- The anchor paper draft, ready for a final submission cycle (target venue: *J. Comput. Phys.* or *Nature Communications*; final venue choice based on Phase 4 results).
- All ablation results documented as a publishable internal technical report and as Sections 5 and 7 of the anchor paper.

Next-semester focus. Case 3 (pitching airfoils with Sharma) is the natural Paper 2 anchored by the existing collaboration on dynamic-stall criteria (Sudharsan et al., 2022; Sharma & Visbal, 2019). Case 2 (electroconvection) is the parallel track if capacity allows. The full Rayleigh–Bénard / spiral-defect-chaos extension with the Mukherjee group is a stretch goal and not on the critical path.

Specific Sharma-group ramp-up. Joint meeting in Week 12–13 to align on Paper 2 scope: dataset format and metadata, target operating-condition sweep, division of authorship, and timeline for a Spring submission. Undergrad B begins reading the Sharma–Visbal LES literature in Week 14 in preparation for Paper 2 ownership.

Week-1 kick-off agenda (suggested)

1. Round-table introductions (10 min): each team member introduces themselves, prior background, and one technical question they want answered by Week 15.
2. Walk-through of the anchor paper (45 min): PI presents the four-page summary; postdoc and undergrads ask clarifying questions.
3. Walk-through of this project plan (30 min): postdoc presents; team discusses risks, role assignments, and whether the role split feels right.
4. Toolchain setup (45 min): GitHub repo creation, branching strategy, GPU access, Slack/email channel, weekly meeting times.
5. Action items and Week 2 deliverables (20 min): each team member leaves with a concrete two-page goal list for Week 2.

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